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Roughness Effects on Flow and Heat Transfer for Additively Manufactured Channels

Recent technological advances in the field of additive manufacturing (AM), particularly with direct metal laser sintering (DMLS), have increased the potential for building gas turbine components with AM. Using the DMLS for turbine components broadens the design space and allows for increasingly small and complex geometries to be fabricated with little increase in time or cost. Challenges arise when attempting to evaluate the advantages of the DMLS for specific applications, particularly because of how little is known regarding the effects of surface roughness. This paper presents pressure drop and heat transfer results of flow through small, as produced channels that have been manufactured using the DMLS in an effort to better understand roughness. Ten different coupons made with the DMLS all having multiple rectangular channels were evaluated in this study. Measurements were collected at various flow conditions and reduced to a friction factor and a Nusselt number. Results showed significant augmentation of these parameters compared to smooth channels, particularly with the friction factor for minichannels with small hydraulic diameters. However, augmentation of Nusselt number did not increase proportionally with the augmentation of the friction factor.
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Introduction

The field of AM has, in recent years, seen increased interest in its technology from various industries. A major technological advance that has piqued this interest is the commercialization of machines that can additively manufacture metal parts from various metal alloys. AM methods using metal include selective laser sintering, selective laser melting (SLM), electron beam melting, and DMLS. These technologies enable building metal parts with complex geometries such as small or winding internal passages that cannot be realized with any other method of manufacturing currently available. Additionally, the cost of making a part with AM does not generally increase with increased complexity. Metals of varying compositions, including high nickel or titanium alloys, can be used for fabrication without a significant increase in difficulty compared to standard ferrous or nonferrous metals.

The advantages of AM have caused the aerospace industry to take particular note of this technology for high temperature applications. There are, however, significant limitations on the mechanical strength of parts made with AM. The particle fusion process results in microstructures and porosities that are difficult to predict and have significant impact on the mechanical properties of the part. Despite the limitations, AM could be advantageous for static components that are subjected to high temperatures, such as some of the parts in the combustor or high pressure turbine of a gas turbine engine.

A major challenge that exists when using AM for producing metal parts is the large surface roughness that is an inherent result of the process. Pressure losses and heat transfer in small channels are affected considerably by the roughness of the channel walls. Most, if not all, roughness removal technologies have limitations that preclude them from being used on small, tortuous passages. Understanding AM tolerances and roughness is necessary if this technology is to be used to produce small flow passages in metal parts, as these factors will greatly influence the pressure losses and convective heat transfer. The objective of this study was to evaluate the surface roughness produced by the DMLS process and measure its influence on flow losses and convective heat transfer through rectangular minichannels. Although this paper contains some observations on the effects of build direction and material choice on surface roughness, the focus of this paper was not to characterize how roughness is influenced by machine parameters and build direction.

Review of Literature

Because metal AM technologies have been commercially available for a short time, the body of public literature on this topic is not very broad. Much of the work done to characterize this process has focused on the mechanical properties. Some work has also been performed on quantifying the surface roughness of the DMLS process, which is of particular interest to this study.

Build orientation has been found by many to have a significant impact on the surface finish. Strano et al. [1] studied the roughness as a function of orientation at many different angles between horizontal and vertical of test pieces made from 316L steel powder using SLM. They found that the roughness was fairly constant at a

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value of about $15\text{ }\mu\text{m}$ for angles between horizontal and 45 deg . After this critical angle, roughness decreased steadily to about $13\text{ }\mu\text{m}$ at the vertical. Delgado et al. [2] measured surface roughness of parts manufactured with DMLS from an iron-based material using various machine parameters and two build orientations. They found that roughness is affected by layer thickness, scan speed, and build direction with the latter having the greatest contribution to the roughness. The arithmetic mean roughness, R_a , of a horizontal surface was measured between $5 < R_a < 8\text{ }\mu\text{m}$ which varied within this range as other machine parameters changed. R_a values were two to three times greater on vertical surfaces. Simionelli et al. [3] performed a similar study and observed comparable trends.

Ventola et al. [4] also found that build direction strongly affects surface roughness with DMLS. Roughness was seen to increase dramatically at angled build directions due to the stair stepping effect inherent with a layer-by-layer manufacturing process. Cooper et al. [5] reported some particularly interesting roughness data from the DMLS samples made of titanium. They studied upward and downward-facing surfaces and reported them ordered from smoothest to roughest. As seen in studies performed by other researchers, the order was: (1) horizontal upward-facing surfaces, (2) sloped upward-facing surfaces, and (3) sloped downward-facing surfaces. Supported surfaces were reported to have roughness values approximately an order of magnitude greater than the horizontal upward-facing surfaces. Common roughness values found in literature range between about $5 < R_a < 50\text{ }\mu\text{m}$ and are strong functions of machine parameters and build direction [1–7].

Others have studied the dimensional tolerances and shrinkage effects of the DMLS samples and found that these effects are dependent on material, machine parameters, and geometry [5,8]. Ning et al. [8] found that smaller geometries are more susceptible to dimensional error due to shrinkage. They developed a method to compensate for dimensional inaccuracies by varying the scan speed.

Since flow in rough pipes has been studied for decades, much exists in the literature regarding this topic. However, only a small number of studies examined flow in minichannels, and even fewer looked at channels with high relative roughness. Huang et al. [9] studied flow through pipes with relative roughness (ε/D) values as high as 42%. They observed that as relative roughness exceeds about 7%, the friction factor in the laminar regime is higher than the theoretical value of $64/\text{Re}$, and transition to turbulent flow occurs at a $\text{Re} < 2000$. Dai et al. [10] compiled a large sampling of friction factor data of liquid flow through minichannels. They discovered large variation in the data presented in literature. They also found that channel geometry had little effect on the friction factor and the transition Reynolds number. Additionally, they concluded that channels with a relative roughness greater than 2% have higher friction factors at low Re than those predicted using laminar flow theory.

Jones [11] examined published data regarding flow in rectangular channels and argued that an equivalent or hydraulic diameter, D_h , is not sufficient for determining friction factor from pressure

drop data; other effects caused by the shape of the channel are important. He proposed using a laminar equivalent diameter, D_L , which uses an empirically derived equation based on the aspect ratio of the channel to correct D_h . Using D_L to calculate the friction factors collapsed the results in the laminar regime onto the theoretical $64/\text{Re}$ line. Jones [11] found that evaluating turbulent data with the corrected diameter gives a much better agreement with known correlations for turbulent pipe flow such as the Colebrook equation.

Few studies were found in the literature to have investigated convective heat transfer of specimens made with AM [4,12,13]. Only one of these studies examined the effect roughness has on convective heat transfer, but their study was limited to external convection only [4]. They found that the roughness significantly augments the heat transfer making it a more desirable surface. No studies were found to have investigated internal heat transfer of additively manufactured channels except for a companion study from the same research group as the authors' [14].

The uniqueness of this study is that it presents the characterization of both pressure loss and heat transfer for minichannels constructed using manufacturer recommended parameters for DMLS. To the authors' knowledge, similar work has not been previously published.

Geometric Characterization

Careful characterization of the channel geometry was vital to the success of this study as flow friction and heat transfer results are strong functions of geometric scales. The following is a description of the test coupons used for this study followed by the methodology and results of characterizing the geometric and roughness parameters of the coupons.

Description of Test Coupons. Eleven coupons were fabricated for this study, all of which were 25.4 mm in length and width. A summary of these coupons and their design specifications is given in Table 1. An image of the computer-aided design (CAD) model of the L-2 x coupon is shown in Fig. 1. Only one coupon (Cyl-Al) was made by conventional machining using 6061 aluminum. This coupon contained 14 round channels running along its length that were each reamed to a diameter of $635\text{ }\mu\text{m}$. The Cyl-Al coupon was used to benchmark the experimental facility and data reduction methodology as compared to generally accepted correlations found in the literature. All other coupons in Table 1 were manufactured with rectangular channels using DMLS. Five of the coupons (L-1 x-Co, L-2 x-Co, M-1 x-Co, M-2 x-Co, and S-2 x-Co) were manufactured by a commercial company. The material used to make these five coupons was a powdered cobalt-chrome-molybdenum-based superalloy (CoCr). Each of these five coupons had a row of uniformly sized rectangular channels that varied in width, height, and number of channels. See Table 1 for dimensions.

The remaining five coupons (L-1 x-In, L-2 x-In, M-1 x-In, M-2 x-In, and S-2 x-In) were made using the same specifications

Table 1 Specifications and dimensions of rectangular channels

Coupon name	Channel shape	Material	Number of channels	Design width (μm)	Design height (μm)	Design D_h (μm)	Design S/D_h	Measured width (μm)	Measured height (μm)	Measured D_h (μm)	R_a (μm)	R_a/D_h
Cyl-Al	Round	Al	14	—	—	635	2.20	—	—	635	—	—
L-1x-Co	Rect.	CoCr	24	457	1016	631	1.33	450	1028	626	9.6	0.015
L-2x-Co	Rect.	CoCr	16	914	2032	1261	1.03	898	2036	1246	12.3	0.010
M-1x-Co	Rect.	CoCr	24	305	660	406	2.06	296	694	415	10.3	0.025
M-2x-Co	Rect.	CoCr	16	610	1321	834	1.55	516	1302	739	12.7	0.017
S-2x-Co	Rect.	CoCr	16	610	610	610	2.12	601	626	614	9.5	0.015
L-1x-In	Rect.	Inc 718	24	457	1016	631	1.33	453	1047	632	10.5	0.017
L-2x-In	Rect.	Inc 718	16	914	2032	1261	1.03	919	2083	1275	10.9	0.009
M-1x-In	Rect.	Inc 718	24	305	660	406	2.06	355	690	469	11.9	0.025
M-2x-In	Rect.	Inc 718	16	610	1321	834	1.55	683	1406	920	13.8	0.015
S-2x-In	Rect.	Inc 718	16	610	610	610	2.12	660	669	664	13.3	0.020

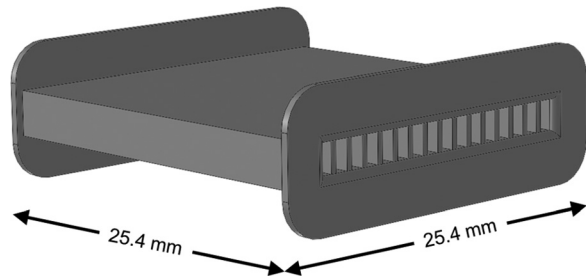


Fig. 1 CAD model rendering of L-2x test coupon

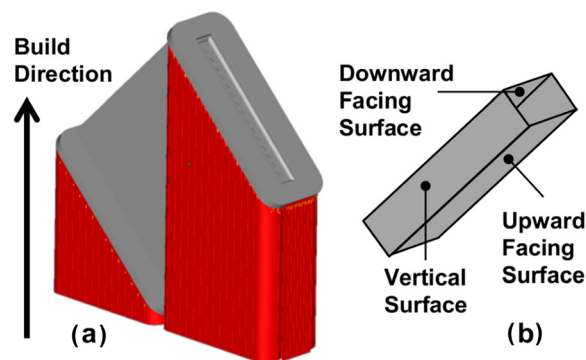


Fig. 2 Image showing (a) build orientation of DMLS coupons with support structures and (b) orientation of channel surfaces relative to build direction

as the five made from CoCr; however, they were made using in-house facilities with a nickel alloy powder that has the same chemical composition as InconelTM 718. Details on the manufacturing of the coupons are given by Snyder et al. [14]. All DMLS coupons in this study were built at a 45 deg angle from the horizontal as illustrated in Fig. 2(a). Support structures used for the coupons fabricated in-house are illustrated in the figure as well. The machine operating parameters used for the laser sintering process were calculated using recommendations from the manufacturer for the specific materials used [14,15].

Characterization of Channel Dimensions. Through the laser sintering process, distortion is a common phenomenon because parts are being subjected to very high thermal gradients during the building process. Warping effects are difficult to predict beforehand and are often determined empirically after a series of test builds. Therefore, the coupons fabricated for these experiments were expected to show some thermal distortion. It was important to quantify the deviation of the coupon geometry from the design specification in order to analyze the pressure drop and heat transfer data using accurate geometric scales.

The manufactured geometry of each coupon was characterized using a commercial X-ray computed tomography (CT) scanner. This instrument was advantageous for geometric measurements because it is nondestructive and gives more accurate results than using traditional measurement tools such as pin gauges and micrometers. The CT scan works by taking successive 2D X-ray images as the test specimen is rotated. The 2D images are then reconstructed into a 3D volume with varying gray-scale gradients that correspond to material density. The resolution of the measurement is dependent on the X-ray power which is adjusted for variations in the material and thickness of the object being examined. Two different power settings were used to examine the coupons which resulted in two different CT scan resolutions. The higher power was used for all Inconel 718 coupons and the L-1 x-Co and S-2 x-Co coupons. The resolution of the 3D images (i.e., voxel

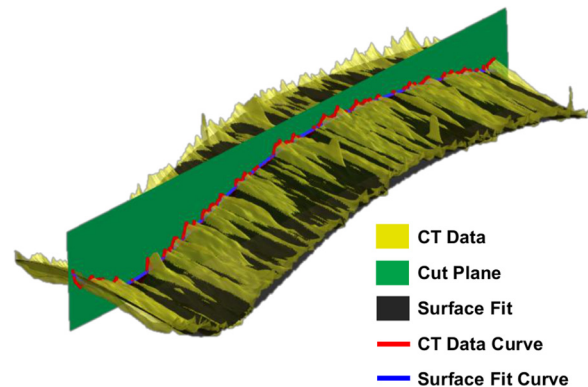


Fig. 3 CT data and fitted surface of the lower surface of one channel of the M-2x-In coupon (note: coordinate axes have different scales)

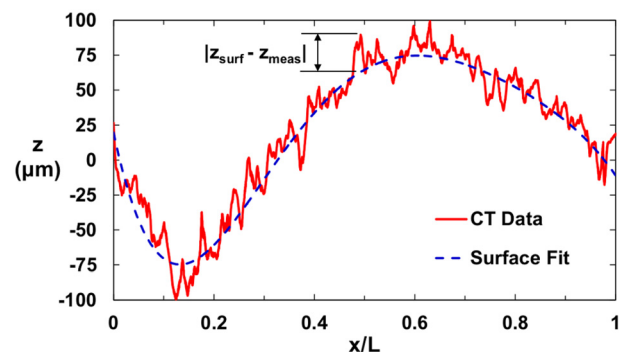


Fig. 4 Comparison of surface fit to CT scan data of the lower surface of one channel of the M-2x-In coupon

size) for these coupons was $35\text{ }\mu\text{m}$. The remaining three CoCr coupons were examined with a lower X-ray power and had a resolution of $23\text{ }\mu\text{m}$.

Using commercial software, the 2D images were combined into a 3D image. This software utilized a sophisticated surface determination algorithm with local adaptation. Accuracies in surface determination of up to 1/10th of the voxel size were achieved with this software [16]. This means that the actual accuracy of the determined surfaces was within $\pm 3.5\text{ }\mu\text{m}$ for the coupons examined with the high-power setting and $\pm 2.3\text{ }\mu\text{m}$ for the coupons examined with the lower power setting. However, this does not suggest that features a few microns in size were resolved. The CT scan was only capable of resolving features as small as the voxel size, but subvoxel precision of the surface location is achieved with the software by interpreting the data between voxels.

After completing the surface determination, a dense point cloud of all the surfaces in the coupon was exported from the analysis software. The point cloud contained about 10×10^6 points for each coupon. An in-house code was developed to take the point cloud data and quantify the dimensions and roughness of each channel surface.

Each surface of every channel was approximated using a polynomial surface fit. The equations of the fit surfaces were third order in the height and width dimensions and fifth order in the streamwise length dimension. The fitting algorithm utilized a bi-square weighting approach to limit the influence of the outliers on the mean surface location. An example of the surface fit to the data is shown in Fig. 3 with an exaggerated scale in the height coordinate to make the variation more visible. To better understand how the surface fits the data, Fig. 4 was generated by cutting the surface fit and CT data using the cut plane illustrated in Fig. 3. The resulting line plots show the deviation of the roughness.

As shown in Fig. 4, the polynomial surface fit expresses the mean surface location. Also indicated in Fig. 4 was the typical warping of the walls of the coupons, which was $\pm 80 \mu\text{m}$.

The fitted surfaces were used to calculate the average manufactured height and width dimensions of each channel. The final height dimension was calculated as the average distance between the top and bottom surfaces, and the final width dimension was calculated in a similar manner using the side surfaces. The average width and height of the channels of each additively manufactured coupon is given in Table 1. The hydraulic diameter, D_h , is also shown for each coupon in the table. Instead of calculating D_h for each channel and averaging those values, it was calculated by summing the cross-sectional areas, A_c , of each channel, summing the perimeters, p , of each channel, and using these values to evaluate D_h . This approach gives a more representative D_h of the whole coupon.

In addition to the design specifications of the coupon geometry, the average measured height and width of each coupon is shown in Table 1. In general, dimensions of the CoCr coupons are close to or smaller than the design specification. The CoCr coupon with the largest manufacturing error is the M-2 x-Co which has a measured D_h that is $95 \mu\text{m}$ smaller than that specified by the design. The CoCr with the smallest error is the M-1 x-Co coupon which has a D_h that is only $9 \mu\text{m}$ larger than the design specification.

The Inconel coupons all measured slightly larger than the design specification with the only exception being the L-1 x-In coupon which had an average channel width that was only $4 \mu\text{m}$ smaller than the specification. Of all the Inconel coupons, the M-2x-In coupon exhibited the greatest deviation from its specification as its D_h was $86 \mu\text{m}$ larger than the specification. The deviation of the average width and height dimensions of all coupons is less than $100 \mu\text{m}$, which is a reasonable tolerance for many manufacturing needs; however, this alters the actual flow area and hydraulic diameter by a relatively significant amount. Variations of this magnitude can have a substantial impact on air flow rate through a minichannel. This was the primary motivation for obtaining the CT scans to characterize the geometry of the coupons.

Characterization of Roughness. Roughness parameters were also calculated from the data collected with the CT machine. As mentioned previously, subvoxel precision of the surface determination can be achieved with the CT machine and analysis software used in this study, but this method is not capable of resolving roughness features smaller than the voxel size. Therefore, many small roughness features are not represented in the scan data and only contribute by “blurring” the surfaces in each CT image.

A digital reconstruction of the CT scan data of the opening of one channel of the M-2x-Co coupon is shown next to an image collected with a light microscope for the same channel in Fig. 5. This figure shows that features on the order of the voxel size ($23 \mu\text{m}$ for CoCr) are visible, but the features smaller than the voxel size are not visible in the CT data rendering. These small features are “smoothed” out during CT scanning.

The roughness parameters of each surface for each channel were determined from the point cloud of data using a computer code that was developed in-house. The code calculated the difference between each data point of the CT scan and the fitted surface. Figure 6 shows a plot of the roughness of two surfaces of one channel of the M-2x-In coupon. The image in Fig. 6(a) is a portion of the surface facing upward at a 45 deg angle during the build (i.e., the normal vector of the surface was pointing upward and angled 45 deg from the build direction vector noted in Fig. 2(b)). This surface is noticeably smoother than the surface facing downward at a 45 deg angle during the build (Fig. 6(b)). Variation in surface roughness has been reported by many to be a strong function of the build orientation [2–4,6]. Generally, the upward-facing surfaces have smaller roughness than downward-facing surfaces.

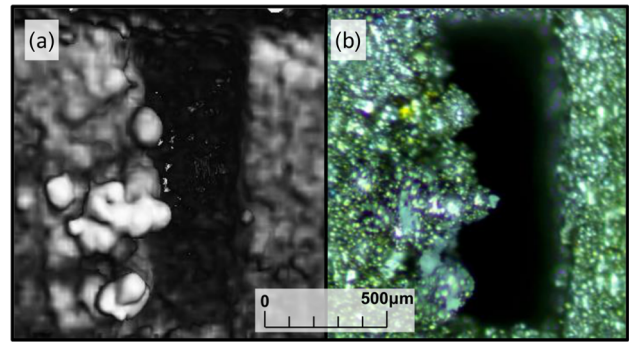


Fig. 5 Image of the opening of a single channel of the M-2x-Co coupon (a) digitally reconstructed from CT scan data and (b) collected with a light microscope

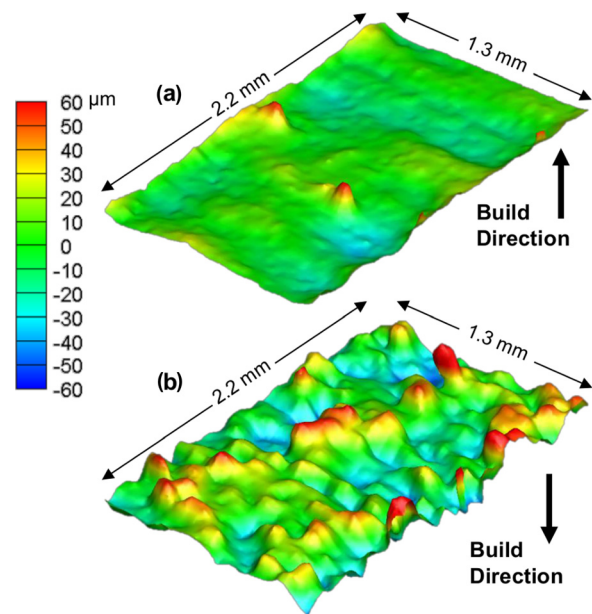


Fig. 6 Contour plots of region from (a) surface facing upward during fabrication and (b) surface facing downward during fabrication of one channel in the M-2x-In coupon

The height differences from the surface fit were used to calculate an arithmetic roughness average, R_a , which is expressed mathematically in Eq. (1). The roughness of each surface in the coupon was calculated using the in-house computer code. The average roughness of each surface in the coupon with the same build orientation is given in Table 2 for all DMLS coupons. The area weighted average roughness of each coupon is also given in Table 2. The measurements suggest that the smoothest surfaces of each coupon are those surfaces facing upward during fabrication with the only exception being the S-2x-Co coupon. This trend agrees with the observation made from Fig. 6 regarding the visible difference between the upward and downward-facing orientations. All other surfaces for a given coupon from the CoCr group have similar roughness values. The resulting roughness for the CoCr coupons with the material and machine parameters used was similar for vertically oriented surfaces as the 45 deg angled downward-facing surfaces.

$$R_a = \frac{1}{m} \sum_{i=1}^m |z_{\text{surf},i} - z_{\text{mean},i}| \quad (1)$$

Contrary to the CoCr coupons, the Inconel coupons exhibited a different roughness for the vertically oriented surfaces than the

Table 2 Roughness of channel surfaces

	R_a , vertical surface (μm)	R_a , 45 deg upward-facing surface (μm)	R_a , 45 deg downward-facing surface (μm)	R_a , area weighted average (μm)
L-1x-Co	8.64	6.93	13.10	9.6
L-2x-Co	14.72	7.06	15.32	12.3
M-1x-Co	11.74	8.59	10.72	10.3
M-2x-Co	14.18	9.22	14.91	12.7
S-2x-Co	10.63	7.90	8.75	9.5
L-1x-In	7.65	10.26	13.14	10.5
L-2x-In	8.82	7.68	15.92	10.9
M-1x-In	7.89	8.25	19.52	11.9
M-2x-In	9.74	8.90	21.90	13.8
S-2x-In	8.97	11.73	23.39	13.3

downward-facing, angled surfaces. In fact, the downward-facing surfaces for a given Inconel coupon had an average roughness that was about twice that of the vertically oriented surfaces. The vertically oriented surfaces had a roughness that compared better to the roughness of the angled upward-facing surfaces than the angled downward-facing surfaces.

Despite the variation in roughness between different surfaces in the same channel, the area weighted average roughness of all coupons was similar in range (9–14 μm). The roughness data presented here is similar in magnitude to the results many have found regarding effects of orientation and overall roughness magnitude [2–4,6].

Experimental Facility

A test rig was built to collect pressure drop and heat transfer measurements of each of the test coupons. The rig (shown in Fig. 7) was built with a smooth contraction chamber that supplied air to the coupon inlet. The inlet contracted the cross-sectional flow area by a factor of 11 for the coupon with the most flow area which ensured a uniform velocity just before the air entered the coupon. The exit expansion chamber was made identical to the inlet for simplicity of fabrication.

Pressure taps were installed upstream of the inlet contraction and downstream of the exit expansion to measure the pressure drop across the coupon. The pressure drop measurement was made with a differential pressure transducer. Gauge pressure downstream of the coupon was measured with a pressure transducer. The mass flow rate of air being delivered to the rig was measured using a laminar flow element (LFE) with accompanying pressure and temperature measurements.

During heat transfer experiments, heat was added to the upper and lower surfaces of the test coupon using electrical resistance surface heaters. The heaters were adhered to copper blocks with a thin layer of thermally conductive paste. Copper blocks were used

because of the high thermal conductivity of copper. The thickness of the copper blocks (25.4 $\times$ 25.4 $\times$ 6.10 mm) was selected to ensure that the temperature along the length of the coupon was uniform to create a constant temperature boundary condition in the coupon. The copper was then fixed to the coupon surface with another layer of thermal paste. Rigid foam blocks were placed adjacent to the heaters to provide insulation to minimize heat loss from the heater and to provide structural support.

The entire assembly was compressed vertically to minimize thermal contact resistance between the various layers. The inlet/exit contraction pieces adjacent to the coupon were made of poly-ether ether ketone which provided mechanical strength to the assembly and has a relatively low thermal conductivity which minimized heat loss. Additionally, the entire assembly was insulated in a layer of foam to further limit heat loss to the environment. Inlet and exit air temperatures were measured with a rake of four thermocouples to get a representative average air temperature.

Several thermocouples were embedded in locations throughout the entire assembly including the copper blocks, the foam blocks, and the plastic contraction pieces. This allowed for determination of the coupon surface temperature as well as calculation of conduction losses through the various components.

Data Analysis Method

Friction factor was calculated using the pressure drop data and Eq. (2). As mentioned above, the pressure drop measurement across the coupon, ΔP , was measured relatively far upstream and downstream where the flow was practically quiescent. The high velocity flow exiting the coupon dumped into the low velocity flow at the exit and dissipated. The losses associated with this dissipation resulted in a pressure drop measurement that was higher than the pressure drop of the air as it traveled through the coupon. To account for the additional losses, ΔP was corrected by assuming the exit loss coefficient was unity

$$f = \Delta P \frac{D_h}{L} \frac{2}{\rho V^2} \quad (2)$$

To perform the heat transfer analysis, the temperature of the coupon surface, T_s , was determined using the temperature measurements in the copper blocks, T_{Cu} , and a 1D conduction analysis through the copper, thermal paste, and coupon wall expressed mathematically in Eq. (3). The thickness variables in Eq. (3) are shown graphically in Fig. 8. The area, A , in the expression is the cross-sectional area of the test stack normal to the heater surface. A similar experimental facility used by Weaver et al. [17] implemented the use of a copper block of the same thickness to create a constant temperature boundary condition. They performed a three-dimensional computational analysis of their setup and found that the variation in temperature in the copper was much less than the increase in temperature of the fluid flowing through the channel which validates the assumption of a constant temperature boundary

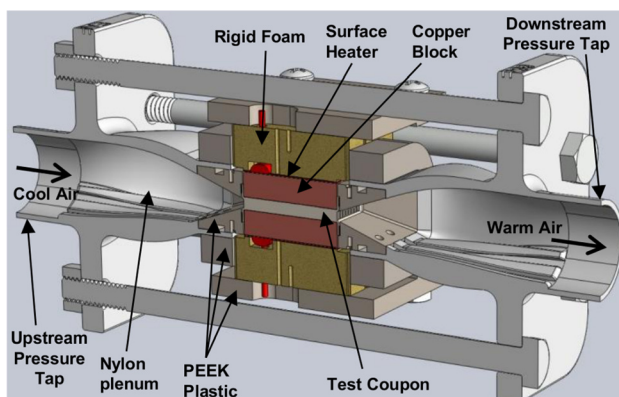


Fig. 7 CAD image of rig used to measure pressure drop and heat transfer in additively manufactured test coupons

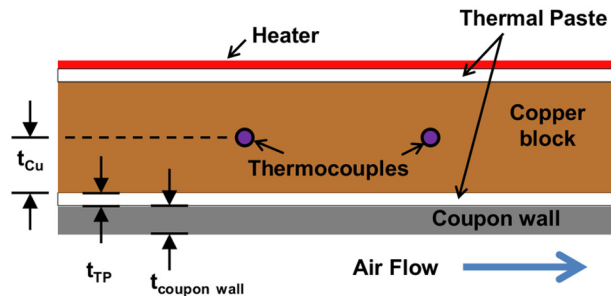


Fig. 8 Cross section view of test stack

$$T_s = T_{Cu} - \frac{Q}{A} \left(\frac{t_{Cu}}{\kappa_{Cu}} + \frac{t_{TP}}{\kappa_{TP}} \frac{t_{wall}}{\kappa_{cou}} \right) \quad (3)$$

The thermal conductivity of the coupon, κ_{cou} , used in Eq. (3) was obtained from the vendor's material specification sheet for those made with CoCr [18]. The vendor does not specify a thermal conductivity for the Inconel 718 but does specify that the parts are fully dense with no significant porosity [19]; the thermal conductivity of these coupons was assumed to be equal to that of the bulk material.

Care was taken during the rig design to minimize thermally conductive paths to reduce the flow of heat into the solid material instead of the fluid. Nonetheless, some degree of conduction through the solid components was unavoidable. The primary heat loss paths were through the rigid foam adjacent to the copper blocks and through the plastic contraction pieces. Conduction losses, Q_{loss} , through these components were calculated using measurements from the embedded thermocouples and a 1D approximation. For two extreme cases, the conductive heat loss was as high as 10%, which occurred for the M-1x-Co coupon at low Reynolds number. The case with the lowest percentage of heat loss was the L-2x-Co coupon at high Reynolds number, which resulted in only a 2% conductive heat loss. The balance was transferred into the fluid.

The convective heat transfer coefficient, h , was calculated using Eq. (4) where A_S is the wetted surface area of the coupon, ΔT_{LM} is the log mean temperature difference as defined in Eq. (5), and Q_{Air} is the heat transferred into the air as it passed through the coupon as determined by Eq. (6). Results of the heat transfer coefficient are presented later in terms of a Nusselt number, Nu ,

$$h = \frac{Q_{Air}}{A_S \Delta T_{LM}} \quad (4)$$

$$\Delta T_{LM} = \frac{T_{Air,2} - T_{Air,1}}{\ln \left(\frac{T_s - T_{Air,1}}{T_s - T_{Air,2}} \right)} \quad (5)$$

$$Q_{Air} = \dot{m} c_p (T_2 - T_1) \quad (6)$$

Q_{Air} should theoretically be equal to the heat added to the system with the electric heaters less the conduction losses. Since the heater power was measured and the conduction losses approximated, there was redundancy in measurements of heat input. An energy balance was calculated for each data point in the heat transfer data set and is presented in Fig. 9. As seen in the figure, the energy balance was within $\pm 10\%$ for all data and was greatest in magnitude at high and low Reynolds numbers.

Measurement Uncertainty

Uncertainty of the measurement method was determined using the analysis described by Figliola and Beasley [20] and the

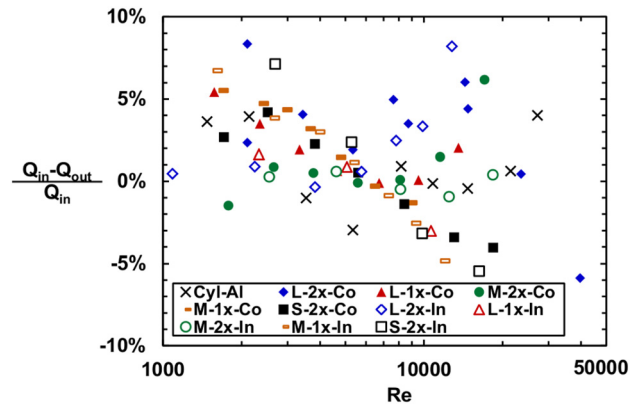


Fig. 9 Energy balance for each test coupon and condition

Table 3 Uncertainty of measurement parameters

Measurement	Uncertainty
P_{up}	± 3.5 kPa
T	± 1 °C
ΔP_{cou}	± 0.55 kPa
ΔP_{LFE}	± 5.5 Pa
W	± 12.7 μ m
H	± 12.7 μ m
L	± 0.13 mm

measurement uncertainty values presented in Table 3. Through a detailed analysis, the primary contributors to the uncertainty of the friction factor were calculated. The largest contributor to the uncertainty was found to vary from coupon to coupon and over the range of Reynolds numbers.

The majority of the friction factor data presented here has an uncertainty between $\pm 6\%$ and 12% . However, the total uncertainties of friction factor for the M-2x-Co, M-2x-In, L-2x-Co, and L-2x-In coupons at low Re approach $\pm 30\%$ which is the highest of any friction factor data presented here. Uncertainty values quickly decreases with increasing Re. The percent contributions to the uncertainty of the friction factor measurements for two extreme conditions are shown in Fig. 10. For the M-2x-Co coupon at low Re, the pressure drop measurement across the coupon is the dominant contributor to the measurement uncertainty because the low Re data is collected close to the measurement limit of the differential pressure transducer.

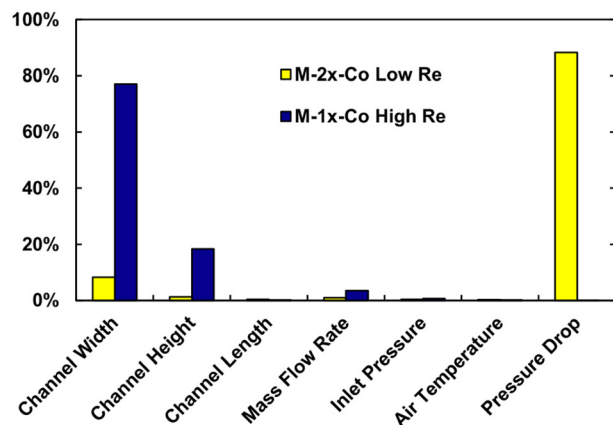


Fig. 10 Percent contribution to uncertainty values of friction factor for two extreme cases

High Re data for the coupons with small geometry (S-2x-Co, S-2x-In, M-1x-Co, M-2x-In, L-1x-Co, and L-1x-In) have uncertainties near $\pm 20\%$. Also shown in Fig. 10 is that for high Re with the M-1x-Co coupon the uncertainty in the friction factor is governed by the uncertainty in the measurement of geometric parameters, which is why it was important to complete the CT scans of the coupons. Uncertainties of the surface determination method are constant for all coupons for the same material. Therefore, the coupons with the smaller geometries, such as the M-1x-Co coupon, have higher relative uncertainties of the geometric measurements. In addition, as the Reynolds number increases, the influence of the velocity on the friction factor increases. Since the bulk velocity calculation is dependent on the cross-sectional flow area, the friction factor equation is highly sensitive to changes in geometry. In fact, it can be shown that the friction factor is proportional to $2 \times H^3 W^3 / (H + W)$, which scales with D_h^5 . For this reason, too, it was important to characterize the geometry in order to reliably reduce pressure drop measurements to friction factors.

A similar analysis was performed to examine the uncertainty of the heat transfer data. In general, the relative uncertainty of Nu for all data ranges between $\pm 2\%$ and 14% . The percent contribution to the uncertainty of the Nusselt number of the measurements involved in the heat transfer experiments are shown in Fig. 11 for two extreme cases. Figure 11 demonstrates the importance of the channel geometry for the S-2x-In coupon and the dominance of the inlet/exit temperatures for the L-2x-In coupon.

Pressure Loss Results

The experimental apparatus was benchmarked using a coupon with smooth, cylindrical holes (Cyl-Al). Friction factor results for this coupon is shown in Fig. 12 compared to laminar theory ($f = 64/Re$) and the Colebrook formula for turbulent flow as defined in Eq. (7) where $\epsilon/D = 0$ for a smooth coupon [21]

$$\frac{1}{\sqrt{f}} = -2.0 \log \left(\frac{\epsilon/D}{3.7} + \frac{2.51}{Re\sqrt{f}} \right) \quad (7)$$

As is seen in Fig. 12 for the smooth coupon (Cyl-Al), the results are in good agreement for the range of Reynolds numbers. However, the data diverges from the laminar theory for $700 < Re < 2000$. The increased friction in this range is the result of entrance effects becoming significant given the coupons were between $20 < L/D_h < 63$. Langhaar [22] quantified entrance effects of channel flow to determine an apparent friction factor for channels of varying length to diameter ratios. Laminar data collected from the Cyl-Al coupons is shown relative to Langhaar's results in Fig. 13, which agrees well for the Cyl-Al coupon.

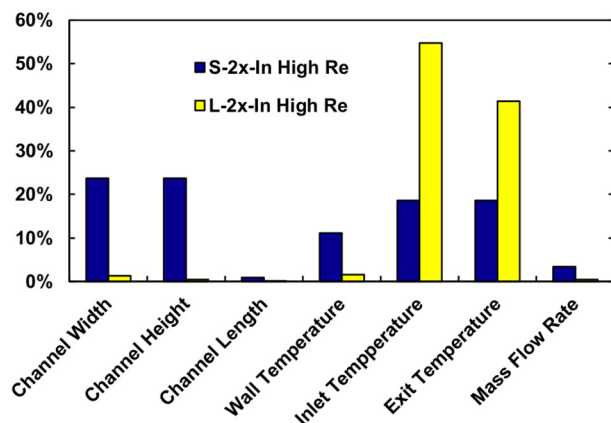


Fig. 11 Percent contribution to uncertainty values of Nusselt number for two extreme cases

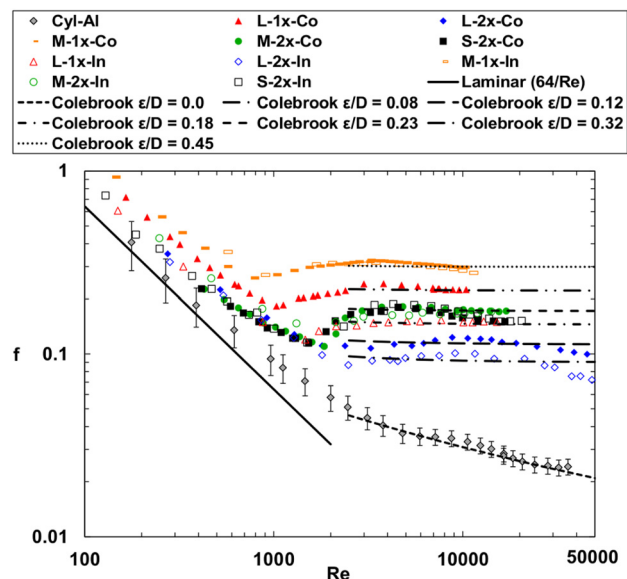


Fig. 12 Friction factor of rectangular channel DMLS coupons

Length-to-diameter ratios do not have such an impact on pressure loss in the turbulent flow region because turbulent flow requires a smaller length to diameter ratio to become fully developed. The agreement of the turbulent data and the correlation in Fig. 12 shows that entrance effects do not contribute significantly to the pressure drop for the range of Reynolds numbers tested here.

The pressure loss data for all DMLS coupons are shown alongside the smooth channel results in Fig. 12. All coupons except the L-1x-Co, M-1x-Co, and M-1x-In have the same friction factor at a given Reynolds number for $Re < 1500$. However, all friction factors for this low range of Reynolds numbers are higher than what is expected from laminar flow theory, which predicts that $f = 64/Re$ regardless of roughness. Much of the experimental work examined by Moody [23] when developing his friction factor chart did not contain data for high relative roughness values [24]. His original paper only presents data for ϵ/D as high as 5%. It is also important to note that the absolute roughness, ϵ , used in Fig. 12 is not equivalent to the arithmetic average roughness, R_a . The absolute roughness can be considered an apparent roughness that is felt by the fluid according to Moody's original data which used sand grain diameter as the absolute roughness value.

Deviation from laminar flow theory of flow through channels with high ϵ/D was seen in the results of several others who have studied friction factor of minichannels [9,10]. The friction factor of the L-1x-Co, M-1x-Co, and M-1x-In coupons in the laminar regime are higher than the other coupons because they have the

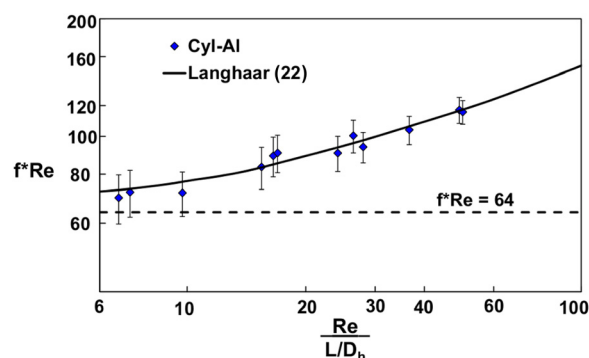


Fig. 13 Friction factor results showing entrance region effects in laminar region for Cyl-Al coupon plotted against data from Langhaar [22]

smallest hydraulic diameter and therefore largest roughness-to-hydraulic diameter ratio, as given in Table 1. The larger relative roughness also reduces the transitional Re where the flow becomes turbulent as observed in Fig. 12. This corroborates the study of mini- and microchannels with large relative roughness done by Huang et al. [9].

In the turbulent region shown in Fig. 12, the data can be categorized into four groups. The first contains the M-1x-Co and M-1x-In coupons, which have a much higher friction factor than the other coupons because they have the smallest diameter of all the coupons and thereby the largest ratio of roughness to hydraulic diameter. A value for ε/D was calculated from the Colebrook equation by curve fitting each coupon's data with the correlation. These relative roughness values for each coupon are given in Fig. 12 along with a line indicating the value.

The L-2x-Co and L-2x-In coupons have the lowest friction factor of all DMLS coupons. These two coupons also have the largest D_h of all coupons. The result of a large D_h is a lower ε/D and lower friction factor than the other coupons. The friction factor of the L-2x-In coupon is slightly lower than the L-2x-Co coupon which is the result of the higher roughness in the L-2x-Co coupon. ε/D values were also estimated for these coupons as shown in Fig. 12. The L-2x-Co has a relative roughness of about $\varepsilon/D = 0.12$ and the L-2x-In is about $\varepsilon/D = 0.08$.

The third group of coupons consists of the M-2x-In, M-2x-Co, S-2x-In, S-2x-Co, and L-1x-In coupons. The friction factor of all these data is similar throughout the whole range of Re. Since these coupons all have a similar D_h , this suggests that the roughness-to-hydraulic diameter ratios are also approximately the same. The relative roughness for these coupons is approximated as $\varepsilon/D = 0.20$.

The L-1x-Co friction factor data falls below the M-1x-Co and M-1x-In data and is in a category of its own. This coupon has a D_h that is similar to those of the third group. The higher friction factor of the L-1x-Co coupon over that of the coupons in group three suggests that it has greater roughness. However, this was not observed in the measured roughness results presented in Table 2. The reason for this discrepancy may be abnormalities in the manufacturing process. The relative roughness as estimated from the plot is $\varepsilon/D = 0.32$.

An absolute roughness, ε , for each coupon was calculated using D_h and the estimated relative roughness based on the Colebrook equation for each coupon is listed in Fig. 12. All of the coupons have an absolute roughness between $100 < \varepsilon < 210 \mu\text{m}$. This value is approximately an order of magnitude higher than the measured arithmetic mean roughness, R_a , as shown in Table 2.

Heat Transfer Results

Convective efficiency, η_c , of the walls separating each of the channels was calculated by treating the separating walls as extended surfaces from the upper and lower walls of the coupon. The convective efficiency is a ratio of the actual heat transfer from an extended surface to the maximum heat transfer from the same surface if it were at a constant temperature [25]. High convective efficiencies are necessary to validate the constant temperature boundary condition assumption used in the data reduction method. Figure 14 shows that the convective efficiency of all coupons tested is greater than 77%. However, the coupons with the lowest efficiency are the L-2x-Co and L-2x-In because the walls are thin and have the greatest height. All other coupons have an efficiency of 90% or greater. Because of the high efficiencies, assuming a constant wall temperature is an acceptable approximation.

Heat transfer data collected for the Cyl-Al coupons is presented in the form of a Nusselt number in Fig. 14. Along with these data, the Dittus–Boelter correlation given by Eq. (8) and Gnielinski correlation given by Eq. (9) are plotted where f in the Gnielinski correlation is taken to be the smooth friction factor found using the Colebrook equation listed above [26,27]. The Dittus–Boelter

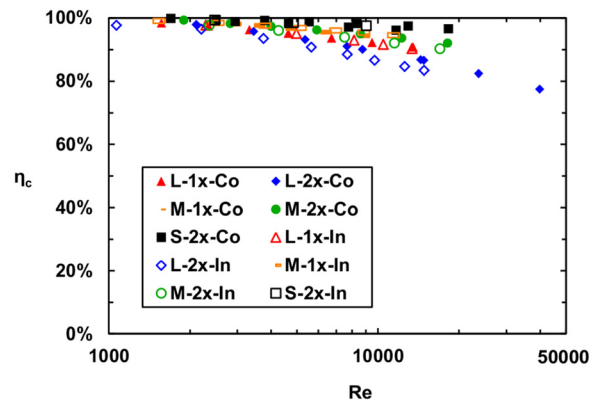


Fig. 14 Channel wall convective efficiencies

equation is only valid for $\text{Re} \geq 10,000$, and the Gnielinski correlation is valid for $3000 \leq \text{Re} \leq 5,000,000$. Within their respective Reynolds number ranges, the data is within $\pm 12\%$ of both correlations. The data in Fig. 15 at low Re appears to be transitioning to a constant value of 3.66 as would be expected for data in the laminar regime

$$\text{Nu} = 0.023 \text{Re}^{0.8} \text{Pr}^{0.4} \quad (8)$$

$$\text{Nu} = \frac{f/8(\text{Re} - 1000)\text{Pr}}{1 + 12.7\sqrt{f/8}(\text{Pr}^{2/3} - 1)} \quad (9)$$

Results of the Nusselt number of the DMLS coupons are given in Fig. 15 with the Cyl-Al heat transfer data. The ordering of the data in this figure from lowest Nu to highest corresponds to the ordering of the friction factor data presented in Fig. 12. Norris [28], as reported by Kays et al. [29], experimented with pressure loss and heat transfer of flow through channels with varying roughness and developed a simplified correlation for Nusselt number augmentation expressed as a function of friction factor augmentation. He also found that the augmentation of Nusselt number increases with increased friction factor according to the functional relationship given in Eq. (10), where $n = 0.68\text{Pr}^{0.215}$.

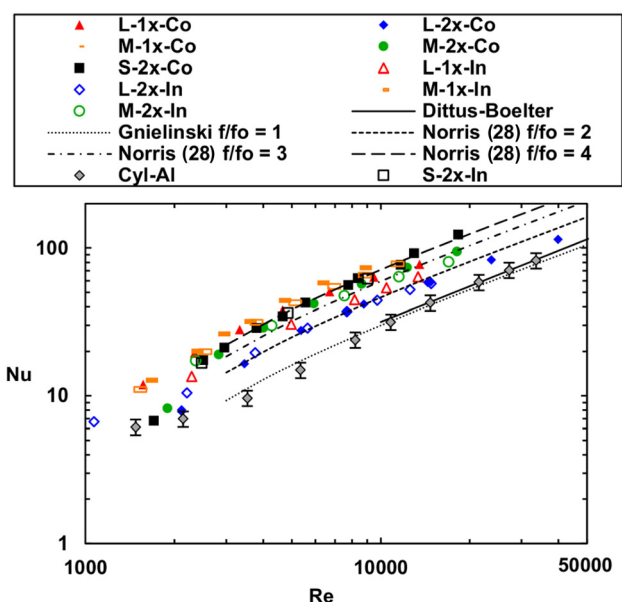


Fig. 15 Nusselt number of rectangular channel DMLS coupons

$$\frac{Nu}{Nu_0} = \left(\frac{f}{f_0} \right)^n \quad (10)$$

Lines of Nu/Nu_0 for varying fff_0 using the above correlation are given in Fig. 15 for the purpose of comparison. As a limitation to this correlation, Norris reported that augmentation of Nu ceases to increase for friction factor augmentation greater than 4.0. All the data in Fig. 15 falls below Norris's correlation where $fff_0 = 4$ except the M-1x-Co and M-1x-In coupons which are slightly above. This difference is not significant because of experimental uncertainty in the data and uncertainty in the correlation.

Increases in pressure loss through channels in gas turbines can result in decreased efficiencies of the engine. Sometimes, however, heat transfer increases may be desirable for a given flow channel despite the pressure losses induced. The augmentation plot shown in Fig. 16 can prove useful for examining the tradeoffs between pressure loss and heat transfer. This figure shows the augmentation of the measured friction factor compared to a smooth channel at a given Re plotted against the augmentation of the measured Nu compared to Nu of a smooth channel at the same Re . The M-1x-Co and M-1x-In coupons have the greatest augmentation of friction factor and Nusselt number, but as noted previously, the augmentation of Nu does not increase proportionally with friction factor for high fff_0 .

Also shown in Fig. 16 is data from Saha and Acharya [30] who studied heat transfer in channels with angled grooves. These angled grooves were made up of a linear array of depressions in the wall of a flat plate. The grooves were all oriented at an angle of 45 deg from the direction of the flow. Four different configurations were studied: small pitch, large pitch, grooves with angled ribs, and grooves with straight ribs. These configurations were found to have a thermal performance similar to or better than channels with various configurations of ribs reported in literature. Augmentations of the grooved channel data are comparable to the augmentations of the DMLS channels except the angled groove-angled rib channel has a better heat transfer augmentation for a given friction factor augmentation than DMLS channels.

As can be gleaned from the data in Fig. 12, fff_0 increases with increasing Re for the DMLS channels for Re above 1000. This increase was also observed by Saha and Acharya in the grooved channel data [30] presented in Fig. 16. This figure shows that Nu/Nu_0 of the grooved channels is fairly constant over the range of Re tested for the grooved channels. However, fff_0 is a strong

function of Re for these channels. This results in a higher cost in pressure loss with little or no increase in heat transfer coefficient augmentation as Re increases. The AM channels studied here show similar trends in that fff_0 is a stronger function of Re than Nu/Nu_0 . However, the heat transfer augmentation decreases at a greater rate than the grooved channel data suggesting the thermal performance decays at a greater rate with increasing Re .

Channel flow in gas turbines is driven by fixed pressure ratios. An analysis was performed to examine the reduction in mass flow rate through M-1x-Co coupon for a fixed pressure ratio compared to a theoretical smooth coupon with the same dimensions and the same pressure ratio. Mass flow through the smooth coupon was calculated using the given pressure ratio from the data and a smooth friction factor correlation. For a given pressure ratio, the DMLS coupon had a 70% reduction in flow rate compared to a smooth channel of the same size. This is due to the very high augmentation of friction factor for the M-1x-Co coupon as shown in Fig. 16. While the mass flow rate decreased significantly, the convective heat transfer rate decreased, but only by 55% for the DMLS coupon compared to the smooth coupon. This shows that the relative reduction in heat transfer is less than the relative reduction in mass flow rate for a fixed pressure ratio. The augmentation of the heat transfer coefficient due to the roughness compensates somewhat for decreases in the heat transfer from the reduced mass flow relative to a smooth channel.

Conclusions

DMLS is a technology that shows much promise for use in the aerospace industry. Understanding the fluid dynamics and heat transfer characteristics of minichannels made with DMLS is necessary for successfully utilizing this manufacturing process for certain applications. The study presented in this paper has examined the roughness due to the DMLS process and its effects on pressure loss and heat transfer in rectangular minichannels.

Using the settings suggested by manufacturers of the DMLS machines, geometric tolerances for minichannels were not met relative to the design intent. The research performed in this study indicated the need to obtain accurate dimensions of the coupons which was achieved with CT scans; the scans also gave results of surface roughness levels. An in-house method was developed to analyze the scans that account for the slight warping of the coupons to determine the minichannel dimensions. Because pressure loss through a channel is a strong function of the geometry, small variations in the hydraulic diameter can have dramatic effects on the friction factor. For this study, deviations from the design hydraulic diameter were as high as one third of that specified for some of the minichannels manufactured.

For the DMLS coupons, the friction factors were significantly increased relative to smooth channels due to the roughness levels. With decreasing hydraulic diameters, the friction factors increased as a consequence of higher roughness-to-hydraulic diameter ratios. While high augmentations resulted in increased pressure losses, the Nusselt number augmentations did not linearly scale with the friction factors. Analysis showed, however, that for a given pressure ratio across the coupons, the higher pressure losses resulted in much lower mass flow and heat transfer rates. However, the reduction in heat transfer was not as large as the reduction in mass flow rate because of the augmentations that occur for the rough DMLS coupons. Comparisons of friction and heat transfer augmentation between AM channels and channels with grooves show that channels made with DMLS have relatively comparable thermal performance. This diminishes the need to design ribs or grooves into walls of small channels for performance enhancement when using DMLS.

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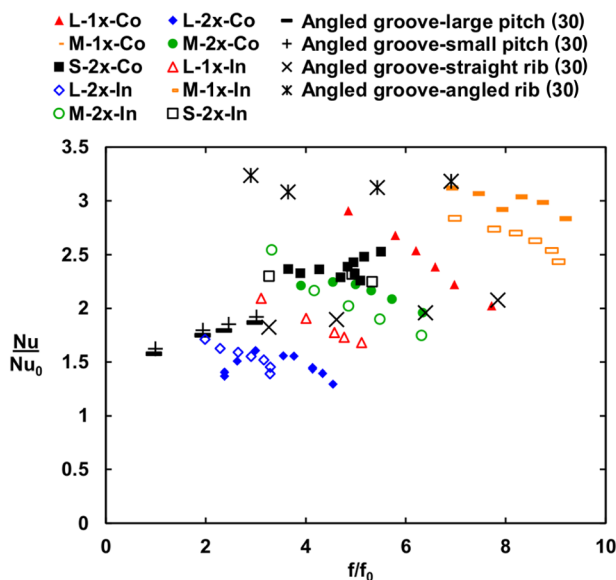


Fig. 16 Heat transfer augmentation versus friction factor augmentation of rectangular channel DMLS coupons

Ted Reutzel, Corey Dickman and Griffin Jones at CIMP-3D is greatly appreciated, as the fabrication of the test coupons and CT scanning would not have been realized without their help.

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Nomenclature

A_c = cross-sectional area
 A_S = wetted surface area
 D = diameter
 D_h = hydraulic diameter; $(4A_c/p)$
 f = Darcy friction factor
 h = convective heat transfer coefficient
 H = channel height
 κ = thermal conductivity
 L = length
 m = number of points
 Nu = Nusselt number; (hD_h/κ_{Air})
 p = perimeter
 P = static pressure
 Pr = Prandtl number
 Q = heat flow rate
 Q_{in} = heat input rate from heaters
 Q_{out} = heat flow rate into air and into rig components
 R_a = arithmetic mean roughness
 Re = Reynolds number; $(\rho V D_h/\mu)$
 S = channel pitch
 t = thickness
 V = fluid velocity
 T = static temperature
 W = channel width
 z = roughness height coordinate
 ΔP = pressure differential
 ΔT_{LM} = log mean temperature difference

Greek Symbols

γ = ratio of specific heats
 ε = absolute roughness (i.e., equivalent sand grain roughness)
 η_c = convective efficiency
 ρ = fluid density

Subscripts

Cou = coupon
 Cu = copper
 LFE = laminar flow element
 S = surface at interface of solid and fluid
 up = upstream of coupon
 wall = coupon wall
 o = reference condition
 1 = inlet
 2 = exit

References

- [1] Strano, G., Hao, L., Everson, R. M., and Evans, K. E., 2013, "Surface Roughness Analysis, Modelling and Prediction in Selective Laser Melting," *J. Mater. Process. Technol.*, **213**(4), pp. 589–597.
- [2] Delgado, J., Ciurana, J., and Rodríguez, C. A., 2012, "Influence of Process Parameters on Part Quality and Mechanical Properties for DMLS and SLM With Iron-Based Materials," *Int. J. Adv. Manuf. Technol.*, **60**(5–8), pp. 601–610.
- [3] Simonelli, M., Tse, Y. Y., and Tuck, C., 2014, "Effect of the Build Orientation on the Mechanical Properties and Fracture Modes of SLM Ti–6Al–4V," *Mater. Sci. Eng. A*, **616**, pp. 1–11.
- [4] Ventola, L., Robotti, F., Dialameh, M., Calignano, F., Manfredi, D., Chiavazzo, E., and Asinari, P., 2014, "Rough Surfaces With Enhanced Heat Transfer for Electronics Cooling by Direct Metal Laser Sintering," *Int. J. Heat Mass Transfer*, **75**, pp. 58–74.
- [5] Cooper, D. E., Stanford, M., Kibble, K. A., and Gibbons, G. J., 2012, "Additive Manufacturing for Product Improvement at Red Bull Technology," *Mater. Des.*, **41**, pp. 226–230.
- [6] Calignano, F., Manfredi, D., Ambrosio, E. P., Iuliano, L., and Fino, P., 2013, "Influence of Process Parameters on Surface Roughness of Aluminum Parts Produced by DMLS," *Int. J. Adv. Manuf. Technol.*, **67**(9–12), pp. 2743–2751.
- [7] Roppenecker, D. B., Grajek, R., Coy, J. A., Irlinger, F., and Lueth, T. C., 2013, "Friction Coefficients and Surface Properties for Laser Sintered Parts," *ASME Paper No. IMECE2013-64549*.
- [8] Ning, Y., Wong, Y. S., Fuh, J. Y. H., and Loh, H. T., 2006, "An Approach to Minimize Build Errors in Direct Metal Laser Sintering," *IEEE Trans. Autom. Sci. Eng.*, **3**(1), pp. 73–80.
- [9] Huang, K., Wan, J. W., Chen, C. X., Li, Y. Q., Mao, D. F., and Zhang, M. Y., 2013, "Experimental Investigation on Friction Factor in Pipes With Large Roughness," *Exp. Therm. Fluid Sci.*, **50**, pp. 147–153.
- [10] Dai, B., Li, M., and Ma, Y., 2014, "Effect of Surface Roughness on Liquid Friction and Transition Characteristics in Micro- and Mini-Channels," *Appl. Therm. Eng.*, **67**(1–2), pp. 283–293.
- [11] Jones, J. O. C., 1976, "An Improvement in the Calculation of Turbulent Friction in Rectangular Ducts," *ASME J. Fluids Eng.*, **98**(2), pp. 173–180.
- [12] Cormier, Y., Dupuis, P., Farjam, A., Corbeil, A., and Jodoin, B., 2014, "Additive Manufacturing of Pyramidal Pin Fins: Height and Fin Density Effects Under Forced Convection," *Int. J. Heat Mass Transfer*, **75**, pp. 235–244.
- [13] Wong, M., Owen, I., Sutcliffe, C. J., and Puri, A., 2009, "Convective Heat Transfer and Pressure Losses Across Novel Heat Sinks Fabricated by Selective Laser Melting," *Int. J. Heat Mass Transfer*, **52**(1–2), pp. 281–288.
- [14] Snyder, J. C., Stimpson, C. K., Thole, K. A., and Mongillo, D., 2015, "Build Direction Effects on Flow and Heat Transfer for Additively Manufactured Channels," *ASME Paper No. GT2015-43935*.
- [15] EOS, 2011, Basic Training EOSINT M 280, Electro Optical Systems GmbH, Munich, Germany.
- [16] Reinhart, C., 2011, "Industrial CT & Precision," Volume Graphics GmbH, Heidelberg, Germany.
- [17] Weaver, S. A., Barringer, M. D., and Thole, K. A., 2011, "Microchannels With Manufacturing Roughness Levels," *ASME J. Turbomach.*, **133**(4), p. 041014.
- [18] EOS, 2011, Material Data Sheet: EOS CobaltChrome MP1, Electro Optical Systems GmbH, Munich, Germany.
- [19] EOS, 2014, Material Data Sheet: EOS NickelAlloy IN718, Electro Optical Systems GmbH, Munich, Germany.
- [20] Figliola, R. S., and Beasley, D. E., 2005, *Theory and Design for Mechanical Measurements*, Wiley, Hoboken, NJ.
- [21] Munson, B. R., Young, D. F., and Okiishi, T. H., 2006, *Fundamentals of Fluid Mechanics*, Wiley, Hoboken, NJ.
- [22] Langhaar, H. L., 1942, "Steady Flow in the Transition Length of a Straight Tube," *ASME J. Appl. Mech.*, **9**, pp. A55–A58.
- [23] Moody, L. F., 1944, "Friction Factor for Pipe Flow," *ASME*, **66**(8), pp. 671–684.
- [24] Nikuradse, J., 1933, "Strömungsgesetze in Rauhen Röhren," *Forschungsheft*, Vol. 4(B), VDI-Verlag, Berlin, p. 316.
- [25] Incropera, F. P., DeWitt, D. P., Bergman, T. L., and Lavine, A. S., 2007, *Fundamentals of Heat and Mass Transfer*, Wiley, Hoboken, NJ.
- [26] Winterton, R. H. S., 1998, "Where Did the Dittus and Boelter Equation Come From?," *Int. J. Heat Mass Transfer*, **41**(4–5), pp. 809–810.
- [27] Gnielinski, V., 1976, "New Equations for Heat and Mass Transfer in Turbulent Pipe and Channel Flow," *Int. Chem. Eng.*, **16**(2), pp. 359–368.
- [28] Norris, R. H., 1971, "Some Simple Approximate Heat Transfer Correlations for Turbulent Flow in Ducts With Surface Roughness," *Augmentation of Convection Heat and Mass Transfer*, American Society of Mechanical Engineers, New York.
- [29] Kays, W. M., Crawford, M. E., and Bernhard, W., 2005, *Convective Heat and Mass Transfer*, McGraw-Hill, Boston.
- [30] Saha, K., and Acharya, S., 2014, "Heat Transfer Enhancement Using Angled Grooves as Turbulence Promoters," *ASME J. Turbomach.*, **136**(8), p. 081004.